Exercise 1: Pointmass-Equations and Forces

A 1-engine propeller aircraft (piston engine) is flying in a windfree standard ISA-atmosphere without a bank angle in a straight flight with a climb angle γ_K of 0.5°. You are given the following data.

GPS-Data:

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-Height	h	5000 m
Climb angle	γ_K	0,5°

Propulsion-Data:

Reference thrust at MSL	T_i	5000 N
Reference density at MSL	ρ_i	$\rho_{s,MSL}$
Reference velocity	V_{ref}	80 kts
Density coefficient	n_ρ	0.75

Aircraft-Data:

Lift coefficient	C_L	1
Zero-lift drag coefficient	$C_{D,0}$	0,012
K-factor	k	0,016
Reference area	S_{ref}	20 m^2
Aircraft mass	m	2000 Kg

In the following you can assume that no aerodynamic side-forces act on the aircraft and that the kinematic bank angle μ_K equals the euler bank angle ϕ.

a) What is the required GPS velocity such that the z-component of the total force in the kinematic frame is zero?
b) Calculate the total force acting on the aircraft. State the result in a suitable coordinate frame.
c) Calculate the current time derivative of the aircraft states.
d) What happens to the time derivatives if the aircraft flies with a kinematic bank angle $\mu_K = 30°$ if all other states remain the same?

Please use the following values in your calculations:

MSL density according to ISA standard atmosphere:	$\rho_{s,MSL} = 1.225 \frac{kg}{m^3}$
MSL pressure according to ISA standard atmosphere:	$p_{s,MSL} = 1.01325e5\ Pa$
MSL temperature according to ISA standard atmosphere:	$T_{s,MSL} = 288.15\ K$
Gravitational acceleration:	$g_0 = 9.80665 \frac{m}{s^2}$
Specific gas constant:	$R = 287.05 \frac{J}{kg\ K}$
Polytropic exponent:	$n = 1.235$
Isentropic exponent:	$\kappa = 1{,}4$
Radius of the earth:	$r_{earth} = 6356766\ \mathrm{m}$